CLINICAL · APPROACH TO

Approach to dizziness and vertigo

"Dizzy" means different things to different patients: spinning, swaying, light-headedness, or a vague sense that the world is not right. The diagnosis rarely comes from the word they use. It comes from the timing and the triggers. Seconds of vertigo on rolling over in bed is BPPV. Hours of vertigo with a full, roaring ear is Ménière's disease, and similar attacks with headache or light sensitivity are vestibular migraine. Days of continuous vertigo is vestibular neuritis, or a stroke. Chronic daily unsteadiness after an earlier vestibular upset is persistent postural-perceptual dizziness. Most dizzy patients can be diagnosed from the history and the eyes, and the one you must not miss is the cerebellar stroke.

AT A GLANCE

- Classify by timing and trigger: episodic triggered, episodic spontaneous, acute continuous, chronic.
- Dix–Hallpike for positional vertigo; HINTS for acute continuous vertigo.
- A normal head impulse in acute vertigo is a red flag for stroke.
- Vestibular migraine is the commonest cause of spontaneous episodic vertigo.

1 CHAPTER ONE The history

Timing and triggers.

1.1 What kind of dizziness?

Ask the patient to describe the sensation without using the word "dizzy". **Vertigo** is an illusion of movement, usually spinning, and points to the vestibular system. **Disequilibrium** is unsteadiness on the feet without a feeling in the head, typical of bilateral vestibular loss, neuropathy and cerebellar disease. **Presyncope** is light-headedness and a sense of impending faint, typical of orthostatic and cardiac causes. **Visual or motion sensitivity**, feeling worse in supermarkets, crowds or scrolling screens, suggests migraine, persistent postural-perceptual dizziness or an uncompensated vestibular loss. Vestibular migraine can produce any of these.

1.2 Timing

TIMING POINTS TO THE CAUSE

PATTERN	TYPICAL DURATION	MOST LIKELY
Episodic, triggered by head position	Seconds (under a minute)	BPPV
Episodic, triggered by pressure, straining or loud sound	Seconds	Superior canal dehiscence, perilymph fistula
Episodic, spontaneous, many brief spells a day	Seconds to a minute	Vestibular paroxysmia
Episodic, spontaneous, with fluctuating hearing, tinnitus and fullness	20 minutes to 12 hours	Ménière's disease
Episodic, spontaneous, with headache, photophobia or visual aura (though not always)	5 minutes to 72 hours	Vestibular migraine
Single acute episode, continuous, with nausea and imbalance	Days, then slowly recovers	Vestibular neuritis (with hearing loss: labyrinthitis); posterior circulation stroke
Transient attacks with other brainstem symptoms	Minutes	Vertebrobasilar TIA
Chronic, most days for more than three months	Persistent	Persistent postural-perceptual dizziness, uncompensated or bilateral vestibular loss, multisensory imbalance of the elderly

1.3 Triggers and associations

Ask what brings the dizziness on. Rolling over in bed, looking up or bending down suggests BPPV. Standing up suggests orthostatic hypotension. Straining, lifting, coughing or loud noise suggests a third window. Sleep loss, stress, caffeine, menstruation and bright light suggest migraine.

Then ask about the ear and the brain. Hearing loss, tinnitus and fullness put the labyrinth at the centre, and fluctuating hearing suggests Ménière's disease. But

acute vertigo with sudden hearing loss can also be an anterior inferior cerebellar artery stroke, because that artery supplies both the labyrinth and the lateral pons. Headache, double vision, dysarthria, dysphagia, numbness, weakness and falls point to a central cause. Ask about migraine past and present, motion sickness as a child, anxiety, ototoxic drugs (gentamicin), head injury, recent ear surgery and barotrauma.

CENTRAL RED FLAGS IN THE DIZZY PATIENT

- Cannot stand or walk unaided during an attack.
- New headache, especially occipital, or neck pain (vertebral dissection).
- Diplopia, dysarthria, dysphagia, facial numbness, limb weakness or ataxia.
- Direction-changing, vertical or purely torsional nystagmus, or nystagmus not suppressed by fixation.
- A *normal* head impulse test in a patient with acute continuous vertigo and nystagmus.
- Vascular risk factors with acute vertigo and sudden hearing loss.

2 CHAPTER TWO Examination

The eyes tell you where.

2.1 Eyes and head

Look for spontaneous nystagmus with the eyes in the primary position and on gaze to each side, with and without fixation (Frenzel lenses remove fixation). **Peripheral** nystagmus is horizontal-torsional, *direction-fixed* with the fast phase away from the affected ear, increases on gaze towards the fast phase (Alexander's law) and is suppressed by fixation. **Central** nystagmus may change direction with gaze, be purely vertical or torsional, and is not suppressed by fixation.

The **head impulse test** looks at the vestibulo-ocular reflex. The patient fixes on your nose while you turn the head briskly through about 15° to each side. A corrective saccade back to the target after a turn towards one side means that canal is weak. The **test of skew**, alternate covering of each eye, reveals a vertical

refixation in skew deviation, a central sign. Check smooth pursuit and saccades, which are normal in peripheral disease.

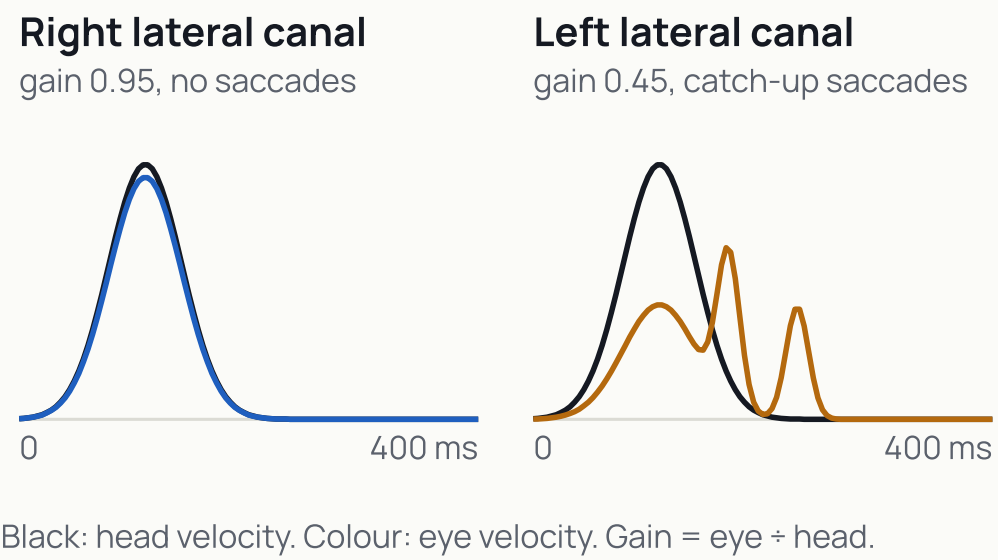


FIGURE 1 The head impulse at the bedside and on video: a weak canal shows low gain and catch-up saccades.

Drawn for ENT Atlas

2.2 Positional testing

The **Dix–Hallpike test** looks for posterior canal BPPV. Turn the head 45° towards the test ear, lie the patient back quickly with the head hanging about 20° below horizontal, and watch the eyes for 30 seconds. A positive test gives upbeating, torsional nystagmus with the upper pole of the eye beating towards the lower ear. It starts after a latency of a few seconds, lasts under a minute, reverses on sitting up and fatigues with repetition. The **supine roll test**, turning the head to each side while lying flat, looks for horizontal canal BPPV, which gives horizontal geotropic or apogeotropic nystagmus.

Positional nystagmus that has no latency, does not fatigue, is purely vertical (especially downbeating), or comes without vertigo suggests a central lesion.

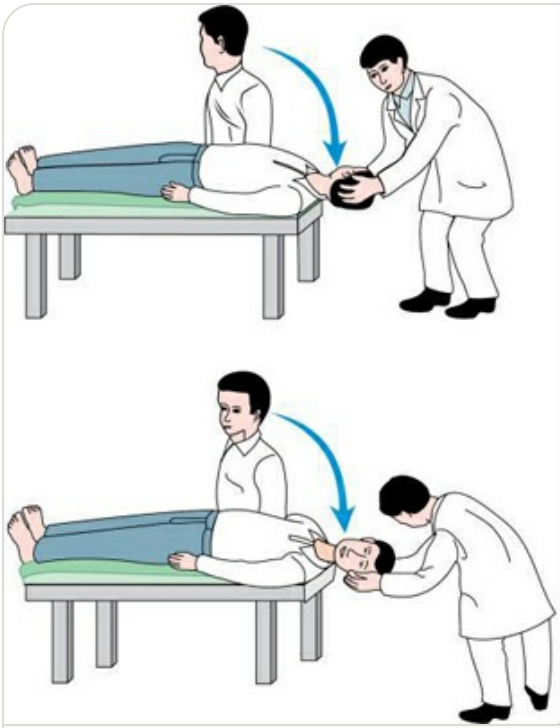


FIGURE 2 The Dix–Hallpike manoeuvre.
Dhingra, fig. 6.1

2.3 The rest of the examination

Otoscopy, tuning forks and a fistula test (pressure on the tragus or with a pneumatic speculum) cover the ear. A cranial nerve and cerebellar examination covers the brain: finger–nose, heel–shin, rapid alternating movements, Romberg and tandem gait. Lying and standing blood pressure looks for orthostatic hypotension, defined as a fall of at least 20 mmHg systolic or 10 mmHg diastolic within three minutes of standing (some definitions allow up to ten); a heart rate rising by 30 or more without hypotension suggests postural tachycardia syndrome. Finally, ask the patient to walk.

3 CHAPTER THREE
Acute continuous vertigo

Ear or brain?

3.1 HINTS

In a patient who has been continuously vertiginous for hours or days, with nausea and spontaneous nystagmus, the question is whether this is vestibular neuritis or a stroke in the posterior circulation, which accounts for roughly one in ten such patients. **HINTS** (Head Impulse, Nystagmus, Test of Skew), done by a trained examiner, is more sensitive for stroke in the first 48 hours than early MRI, which misses small infarcts.

A *peripheral* pattern is reassuring only when all three agree: an *abnormal* head impulse test (a corrective saccade towards the affected side), direction-fixed horizontal nystagmus, and no skew. *Any* central feature (a normal head impulse, direction-changing nystagmus or skew) means stroke until proved otherwise. **HINTS-plus** adds new hearing loss as a central warning, because of anterior inferior cerebellar artery infarcts. Inability to stand unaided is an additional central sign.

HINTS IN ACUTE CONTINUOUS VERTIGO

TEST	PERIPHERAL (NEURITIS)	CENTRAL (STROKE)
Head impulse	Abnormal: catch-up saccade	Normal
Nystagmus	Direction-fixed, horizontal-torsional, suppressed by fixation	Direction-changing on gaze, vertical or purely torsional
Test of skew	No vertical refixation	Vertical refixation (skew)
Plus: hearing	Usually normal (labyrinthitis if lost)	New loss suggests AICA infarct
Gait	Leans but can stand	Cannot stand unaided

3.2 Management

A central pattern needs urgent MRI with diffusion-weighted imaging and the stroke team. A peripheral pattern is managed with short-term vestibular sedatives and antiemetics (prochlorperazine, cyclizine) for no more than a few days, fluids, and early mobilisation with vestibular rehabilitation exercises. A short course of oral steroids may be given early for vestibular neuritis. Sedatives slow compensation if they are continued.

CHAPTER FOUR

4 Recurrent episodic vertigo

The big three.

4.1 BPPV

BPPV is the commonest cause of vertigo: brief spinning on head movement, often first noticed on getting out of bed, typically in older adults and after head injury or a period in bed. The posterior canal is affected in about 85 to 90%. The diagnosis is made by the Dix–Hallpike test, and the cure is a repositioning manoeuvre: the *Epley* (or Semont) manoeuvre for the posterior canal, and the *Lempert (barbecue) roll* or Gufoni manoeuvre for the horizontal canal. Recurrence is common, so patients can be taught self-treatment. Medication has no role, and atypical or refractory cases need MRI.

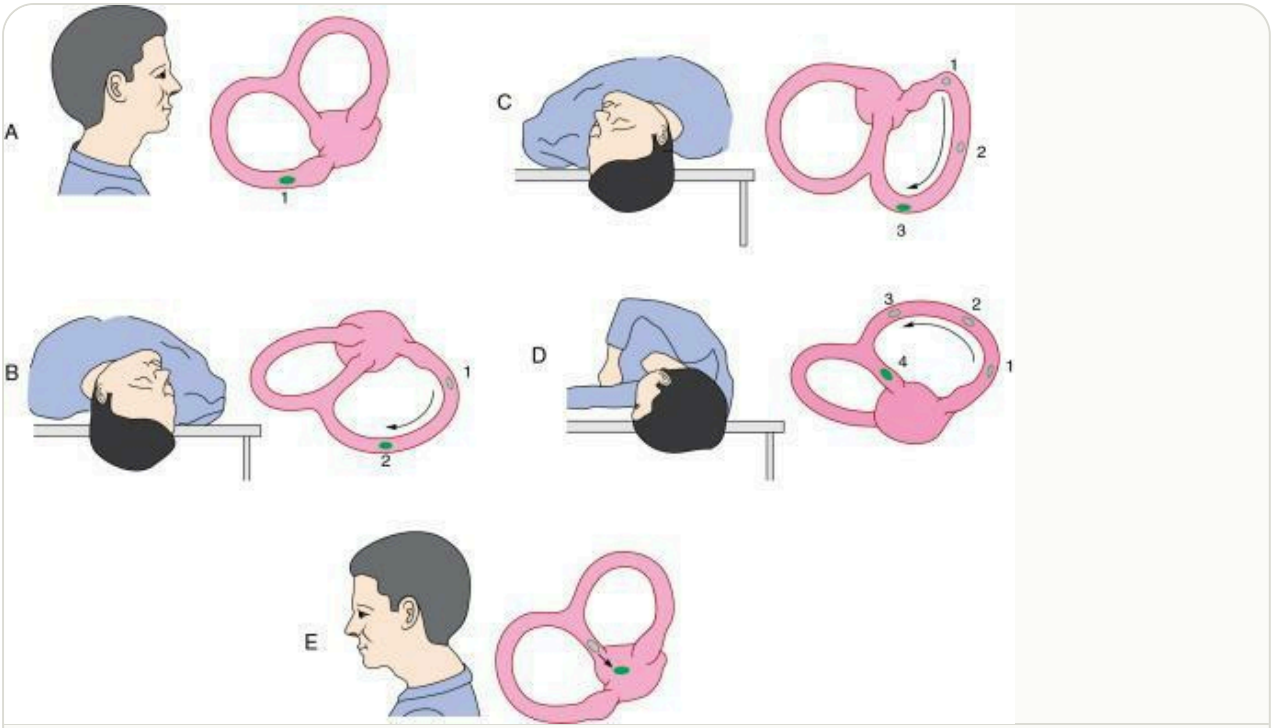


FIGURE 3 The Epley manoeuvre for posterior canal BPPV, with the position of the debris in the canal at each step.

Dhingra, fig. 7.1

4.2 Ménière's disease

Definite Ménière's disease (Bárány Society) requires two or more spontaneous episodes of vertigo, each lasting 20 minutes to 12 hours, an audiometrically documented low- to mid-frequency sensorineural loss in the affected ear on at least one occasion, and fluctuating aural symptoms (hearing, tinnitus or fullness) in that ear, not better explained by another diagnosis. An audiogram is essential, and asymmetric sensorineural loss means MRI.

Treatment aims to reduce the number and severity of attacks. It runs from acute vestibular sedatives through lifestyle measures (moderation of salt, caffeine and alcohol, though the evidence is weak), betahistine (in widespread use despite a large trial showing no benefit over placebo) and diuretics, to intratympanic steroid, intratympanic gentamicin and, rarely, surgery.

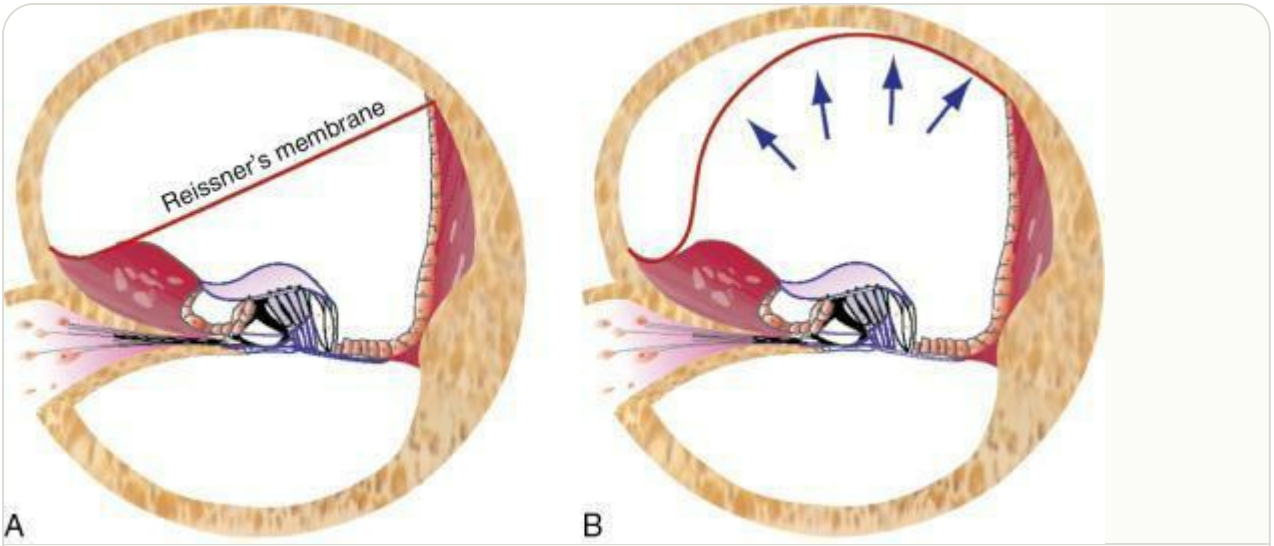


FIGURE 4 (A) The normal cochlear duct. (B) Endolymphatic hydrops: the duct distended and Reissner's membrane bulging.

Dhingra, fig. 15.1

4.3 Vestibular migraine

Vestibular migraine is the commonest cause of spontaneous recurrent vertigo, and it is underdiagnosed because many attacks come without headache. *Definite* vestibular migraine (Bárány Society and International Headache Society) requires at least five episodes of moderate or severe vestibular symptoms lasting 5 minutes to 72 hours. The patient must have current or previous migraine, and at least half the episodes must carry at least one migrainous feature: a migraine-type headache, photophobia with phonophobia, or a visual aura.

There is no confirmatory test, so the diagnosis is clinical. Treatment begins with sleep, regular meals, hydration and the identification of triggers such as caffeine and alcohol. Acute attacks are treated with prochlorperazine and, for headache, a triptan with an NSAID and an antiemetic. Frequent attacks need prophylaxis: propranolol, amitriptyline or nortriptyline, or candesartan. The trial evidence comes from migraine headache rather than vestibular migraine itself, and CGRP antibodies are emerging.

MÉNIÈRE'S OR VESTIBULAR MIGRAINE?

The two overlap more than the textbooks admit, and some patients have both. Documented, progressive low-frequency hearing loss favours Ménière's disease. Headache, photophobia, visual aura, motion sensitivity, a long migraine history and

attacks lasting more than 12 hours or only minutes favour migraine. When in doubt, a trial of migraine prophylaxis is safe and often diagnostic.

4.4 Less common episodic causes

Superior canal dehiscence causes vertigo with straining or loud sound, autophony and pulsatile tinnitus, with low bone conduction thresholds; it is confirmed by VEMPs and CT. **Perilymph fistula** follows stapes surgery, head injury or barotrauma. **Vestibular paroxysmia**, from a vascular loop compressing the eighth nerve, causes many brief spells a day and responds to carbamazepine. **Episodic ataxia type 2** (CACNA1A) is rare and responds to acetazolamide. **Vertebrobasilar TIAs** cause brief vertigo with other brainstem symptoms in patients with vascular risk.

5 CHAPTER FIVE Chronic dizziness

Months of unsteadiness.

5.1 The causes

Chronic dizziness usually has one of five causes, alone or in combination. An *uncompensated unilateral vestibular loss* after neuritis or surgery leaves symptoms on quick head turns. *Bilateral vestibular loss*, from gentamicin, meningitis, bilateral Ménière's disease or idiopathic causes, gives oscillopsia and unsteadiness in the dark. *Persistent postural-perceptual dizziness* is non-spinning dizziness and unsteadiness present on most days for more than three months, worse when upright, moving or in busy visual surroundings; it usually follows a vestibular event, migraine or anxiety, examination and tests are normal, and it is treated with vestibular rehabilitation, cognitive behavioural therapy and SSRIs. The *multisensory imbalance of the elderly* combines vestibular, visual, proprioceptive and musculoskeletal decline with drugs. *Central causes*, cerebellar degeneration, multiple sclerosis and posterior fossa tumours, are rare but must be sought.

Vestibular rehabilitation is the treatment for most chronic vestibular dizziness, and vestibular sedatives should be stopped.

5.2 Medical causes

Many dizzy patients have no ear disease at all. Orthostatic hypotension, arrhythmias and aortic stenosis cause presyncope. Hypoglycaemia, anaemia and thyroid disease, drugs (antihypertensives, anticonvulsants, sedatives), anxiety and panic, and hyperventilation complete the list. Hyperventilation causes dizziness without nystagmus, unless the eighth nerve is demyelinated or compressed.

6 CHAPTER SIX Investigations

When and which.

6.1 Tests that help

A **pure tone audiogram** is needed for every patient with ear symptoms and is required to diagnose Ménière's disease. **MRI of the internal auditory meati and brain** is needed for asymmetric hearing loss or tinnitus, central signs, atypical positional nystagmus and suspected stroke. **Vestibular tests** (videonystagmography, calorics, video head impulse test, VEMPs, rotational chair) quantify and localise vestibular loss; they are useful when the history is unclear, before ablative treatment, and in suspected bilateral loss or dehiscence. **CT temporal bones** is for suspected dehiscence or fistula. An **ECG**, lying and standing blood pressure, glucose and blood count are for presyncope.

THE DIZZY PATIENT

1 Describe the dizziness and its timing

→ **single acute continuous attack**
go to step 2

→ **recurrent episodes**
go to step 3

→ **chronic, most days**
go to step 4

→ **presyncope**
lying and standing BP, ECG, medical review

2 Acute continuous vertigo: do HINTS-plus

→ **any central sign, cannot stand, new headache or neurology**
urgent MRI-DWI and stroke team

→ **all peripheral**
vestibular neuritis — short sedation, early mobilisation, consider steroids

3 Recurrent episodes: what triggers them?

→ **head position, seconds**
Dix–Hallpike, then Epley

→ **spontaneous, 20 minutes to 12 hours, fluctuating hearing**
audiogram, MRI — Ménière's

→ **spontaneous with migrainous features**
vestibular migraine

→ **straining or sound**
CT, VEMP — dehiscence or fistula

4 Chronic dizziness

→ **head-movement related**
uncompensated loss — vestibular rehabilitation

→ **oscillopsia, worse in the dark**
bilateral loss — vestibular tests

→ **visual and motion sensitivity, normal tests**
persistent postural-perceptual dizziness

→ **cerebellar signs**
MRI

6.2 Children

Dizziness in children is uncommon and is usually benign. Vestibular migraine and *benign paroxysmal vertigo of childhood* (brief attacks in toddlers, a migraine precursor) are the commonest causes. Otitis media with effusion, vestibular neuritis, head injury and BPPV after trauma follow, along with the third-window effect of an enlarged vestibular aqueduct. Posterior fossa tumours, seizures and cardiac causes are the rare serious ones to exclude.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Classification by timing and trigger,** not by the word "dizzy".
- **Peripheral versus central nystagmus;** Alexander's law.
- **HINTS and HINTS-plus:** why a normal head impulse is alarming.
- **Dix–Hallpike technique and the Epley manoeuvre;** horizontal canal BPPV.
- **Diagnostic criteria** for Ménière's disease and vestibular migraine.
- **Persistent postural-perceptual dizziness.**
- **When to scan;** the role of vestibular rehabilitation.

7.2 Common pitfalls

COMMON PITFALLS

- Labelling acute vertigo "labyrinthitis" without examining the eyes.
- Prescribing long-term vestibular sedatives, which delay compensation.
- Missing vestibular migraine because the patient has no headache during attacks.

- Giving betahistine for BPPV instead of doing an Epley manoeuvre.

Written for this app from Oxford Textbook of Otolaryngology ch. 7, 14, 15, 85, Oxford Specialist Handbook ch. 3, Oxford Handbook of ENT ch. 3, Dhingra ch. 7, K. J. Lee ch. 16. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

INNER EAR & IAC · TUMOURS

Vestibular Schwannoma

IMAGING FINDINGS

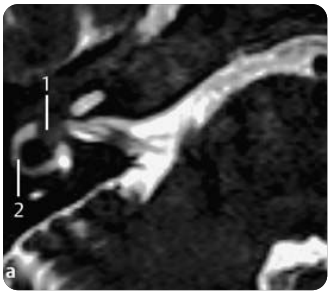
- Enhancing mass centred on the porus, "ice cream on a cone"
- Dark filling defect in bright CSF on high-resolution T2; fundal cap of fluid
- Acute angle with the petrous bone; widened porus; cystic change in larger tumours

TOP DIFFERENTIALS

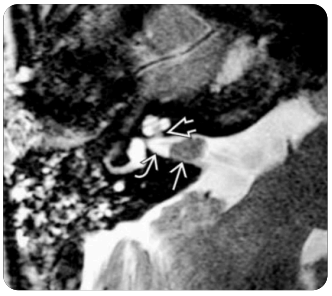
- Meningioma (broad dural base, dural tail)
- Epidermoid cyst (DWI bright)
- Facial nerve schwannoma (labyrinthine segment)

CHECKLIST

- Size in canal and cistern; brainstem and fourth ventricle (Koos)
- Fundal cap and cochlear nerve canal for hearing preservation



MRI, T2-weighted, axial.A partial lack of fluid in the vestibule and horizontal semicircular canals is evident (1). A mass lesion is present in the vestibul…



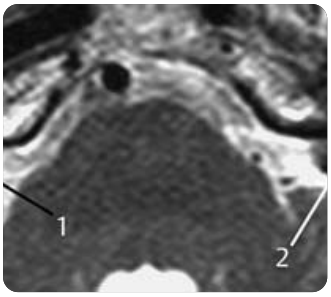
Axial T2WI MR reveals a small intracanalicular vestibular schwannoma visualized as a soft tissue intensity mass surrounded by…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1222.

INNER EAR & IAC · INFECTION & INFLAMMATION

Labyrinthitis and Subclinical Meningitis

- Otoscopy may exclude middle ear infections although CT may be indicated in some cases. MRI will show (retro)cochlear enhancement in cases of (acute) infections.
- Inflammatory involvement of the intralabyrinthine structures may result in fibrosis and ossification. In bilateral cases, this may hamper insertion of a cochlear implant.

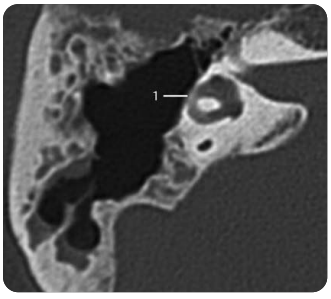


MRI, T2-weighted, axial. Note the normal fluid content of the internal auditory canal on the right side (1), whereas on the affected side, the…

MIDDLE EAR & MASTOID · AFTER SURGERY

Iatrogenic Fenestration

- In fenestrated patients, no actual compression due to masses will be seen.
- Beware of suction of the cavity for cleaning. This must be performed very carefully to prevent severe vertigo and inner ear dysfunction.



CT, axial. Before the availability of microsurgical techniques and the use of ossicular replacement prostheses, mobilization of the footplate was…